

TechNova Solutions Pvt. Ltd.

Enterprise Knowledge Base

AI Department Handbook

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Version History

Version	Date	Author	Changes
1.0	July 2026	AI Department	Initial Release

Table of Contents

- Purpose
- Scope
- AI Department Overview
- Team Structure
- AI Project Lifecycle
- Machine Learning Development
- Deep Learning Development
- Large Language Models (LLMs)
- Retrieval-Augmented Generation (RAG)
- Vector Databases
- Prompt Engineering
- AI Model Evaluation
- Model Deployment
- MLOps Practices

- 15. Responsible AI
 - 16. Approved AI Tools
 - 17. Frequently Asked Questions
 - 18. Related Documents
-

1. Purpose

This handbook establishes the standards, workflows, and best practices followed by the Artificial Intelligence Department. It ensures that AI solutions are accurate, scalable, secure, and aligned with business objectives.

2. Scope

This handbook applies to:

- AI Engineers
 - Machine Learning Engineers
 - Data Scientists
 - Data Engineers
 - NLP Engineers
 - MLOps Engineers
 - AI Interns
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3. AI Department Overview

The AI Department develops intelligent solutions for business automation, predictive analytics, recommendation systems, document intelligence, computer vision, and conversational AI.

Primary responsibilities include:

- Building AI-powered applications
 - Training machine learning models
 - Developing LLM-based assistants
 - Deploying production AI systems
 - Supporting business teams with AI solutions
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4. Team Structure

The department consists of:

- Data Science Team
- Machine Learning Team
- Deep Learning Team
- NLP Team
- Computer Vision Team
- MLOps Team
- AI Research Team

Each team collaborates closely with Engineering and Product teams.

5. AI Project Lifecycle

Every AI project follows these stages:

1. Business Problem Definition
2. Data Collection
3. Data Cleaning
4. Exploratory Data Analysis (EDA)
5. Feature Engineering
6. Model Development
7. Model Evaluation
8. Deployment
9. Monitoring
10. Continuous Improvement

Documentation is required at every stage.

6. Machine Learning Development

Standard workflow:

- Define target variable
- Split training and testing data
- Handle missing values
- Encode categorical features
- Scale numerical features
- Train baseline models

- Optimize hyperparameters
- Evaluate performance
- Deploy approved model

Common algorithms:

- Linear Regression
 - Logistic Regression
 - Decision Tree
 - Random Forest
 - XGBoost
 - Support Vector Machine
 - K-Nearest Neighbors
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7. Deep Learning Development

Deep learning is used for complex tasks such as image classification, speech recognition, and natural language processing.

Frameworks:

- TensorFlow
- Keras
- PyTorch

Common architectures:

- Artificial Neural Networks (ANN)
- Convolutional Neural Networks (CNN)
- Recurrent Neural Networks (RNN)
- LSTM
- GRU
- Transformer Models

GPU acceleration is recommended for training deep learning models.

8. Large Language Models (LLMs)

LLMs are used for:

- Chatbots
- Question Answering

- Text Summarization
- Document Analysis
- Code Generation
- Content Creation

Approved model providers include:

- OpenAI GPT
- Llama
- Mistral
- Gemma
- Phi

Model selection depends on cost, latency, privacy, and accuracy.

9. Retrieval-Augmented Generation (RAG)

The company standard RAG pipeline includes:

1. Document Loading
2. Text Chunking
3. Embedding Generation
4. Vector Storage
5. Similarity Search
6. Context Retrieval
7. LLM Response Generation
8. Response Validation

Benefits:

- Reduces hallucinations
 - Uses company knowledge
 - Provides source-grounded answers
 - Supports enterprise search
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10. Vector Databases

Approved vector databases:

- FAISS

- ChromaDB
- Pinecone
- Weaviate

Selection depends on:

- Dataset size
 - Cloud requirements
 - Scalability
 - Cost
 - Retrieval speed
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11. Prompt Engineering

Guidelines:

- Use clear instructions.
- Define system roles.
- Provide sufficient context.
- Specify output format.
- Avoid ambiguous prompts.
- Test prompts with multiple scenarios.

Prompt templates should be version-controlled.

12. AI Model Evaluation

Evaluation metrics include:

Classification

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

Regression

- MAE
- MSE
- RMSE

- R^2 Score

LLM Evaluation

- Groundedness
- Relevance
- Context Recall
- Faithfulness
- Response Quality

Models must meet minimum performance thresholds before deployment.

13. Model Deployment

Models are deployed through CI/CD pipelines.

Deployment environments:

- Development
- Testing
- Staging
- Production

Every deployment requires:

- Performance validation
 - Security review
 - Rollback plan
 - Monitoring configuration
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14. MLOps Practices

The AI team follows MLOps principles:

- Dataset versioning
- Model versioning
- Experiment tracking
- Automated retraining
- Performance monitoring
- Drift detection
- CI/CD integration

15. Responsible AI

All AI systems must follow these principles:

- Fairness
- Transparency
- Privacy
- Security
- Accountability
- Human oversight

Bias testing must be completed before production deployment.

16. Approved AI Tools

Category	Tools
Programming	Python
ML	Scikit-learn
Deep Learning	TensorFlow, PyTorch
LLM Framework	LangChain
Vector DB	FAISS, ChromaDB
Data Analysis	Pandas, NumPy
Visualization	Matplotlib, Power BI
Deployment	Docker, Kubernetes

17. Frequently Asked Questions

Which framework is recommended for traditional machine learning?

Scikit-learn.

Which framework is used for LLM applications?

LangChain.

Which vector database is recommended for local development?

FAISS.

Why does the company use RAG?

To improve answer accuracy by retrieving relevant company knowledge before generating responses.

How are AI models monitored after deployment?

Performance metrics, prediction quality, latency, and model drift are continuously monitored.

AI Technology Summary

Technology	Standard
Programming Language	Python
Machine Learning	Scikit-learn
Deep Learning	TensorFlow / PyTorch
LLM Framework	LangChain
Vector Database	FAISS
Deployment	Docker & Kubernetes
Experiment Tracking	MLflow
Monitoring	Prometheus & Grafana

Related Documents

- [Engineering Handbook](#)
- [Information Security Handbook](#)
- [Product Documentation](#)
- [Troubleshooting Guide](#)